Developer Setup & Architecture Guide

Overview

The Drug Development Portfolio Dashboard is an enterprise-grade clinical research portfolio management system designed for pharmaceutical organizations to track, manage, and analyze their drug development programs across different therapeutic areas and development phases.

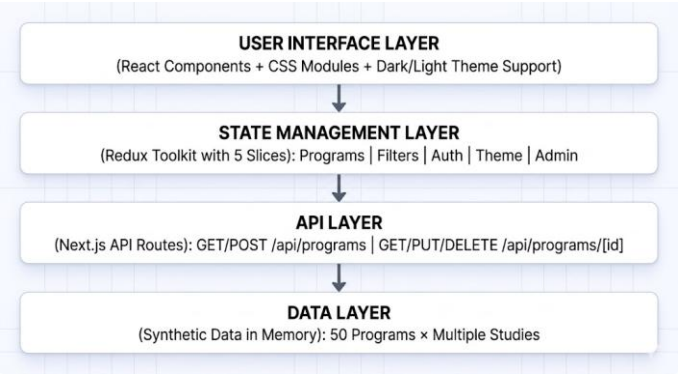
Key Objectives

- Centralized view of all drug development programs
- Real-time tracking of clinical studies and enrollment data
- Role-based access control for data security
- Development timeline visualization with progress tracking
- Comprehensive audit logging for compliance

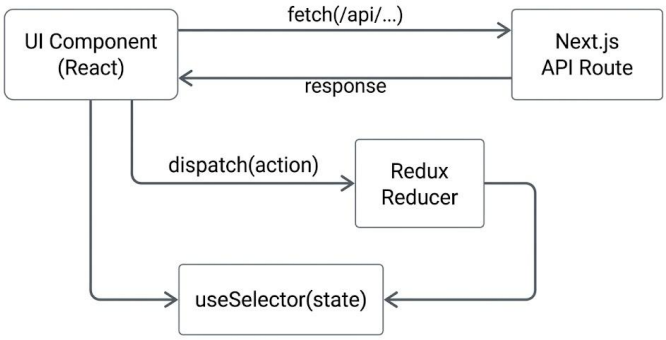
Target Users

- Portfolio Managers: Strategic overview of all programs
- Clinical Program Managers: Detailed program tracking and updates
- Administrators: System management and user oversight
- Data Analysts: Enrollment and milestone tracking

Architecture



Data Flow Diagram



Core Features

- Program browsing and filtering
- Program details with timeline
- Study enrolment tracking
- Program editing
- Program deletion (admin)
- Dark/Light theming

Admin Features

- User management
- Role assignment
- Audit logging
- Activity tracking

Security Features

- Role-based access control
- Feature-level permissions
- Audit trail
- User session management

Design Patterns

- Redux Slices: Organized state management by domain
- Async Thunks: Centralized API call handling
- Custom Hooks: Reusable selector logic
- Component Composition: Modular, testable components
- CSS-in-JS Variables: Dynamic theming system

Technology Stack

Frontend

- Next.js 15.5.14 - React framework with App Router
- React 18.2.0 - UI library
- TypeScript - Type safety and developer experience
- Redux Toolkit 1.9.7 - Predictable state management
- React-Redux 8.1.3 - React bindings for Redux

Browser Support

- Chrome/Edge 90+
- Firefox 88+
- Safari 14+
- Mobile browsers (iOS Safari, Chrome Android)